

Stack Overflow Developer Survey Analysis Technology Trends in Programming Languages and Databases

Part 1



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OUTLINE



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EXECUTIVE SUMMARY



- Analysis of the Stack Overflow Developer Survey dataset to identify technology trends among developers.
- The study focuses on programming languages, databases, and technology preferences.
 - Identification of the most commonly used technologies.
 - Comparison between current technologies and those developers want to learn next year
 - Visualization of trends using charts and dashboards
- JavaScript, SQL and HTML/CSS remain the most widely used programming languages.
- PostgreSQL and MySQL dominate among databases.
- Interactive dashboards were created to explore technology trends.



INTRODUCTION



- Technology trends change rapidly in the software development industry.
- Understanding these trends helps developers and companies make better decisions.
- The Stack Overflow Developer Survey provides valuable data about developer tools and preferences.
- This project analyzes the survey data to identify patterns in technology adoption.
 - Programming languages used by developers.
 - Database technologies and future learning interests.



METHODOLOGY



- The dataset was obtained from the Stack Overflow Developer Survey.
- Data preprocessing was performed using Python and pandas.
- Missing values were handled and multi-value columns were split into separate rows.
- Top technologies were identified using frequency analysis
 - Data visualization was performed using charts
 - An interactive dashboard was created using Cognos Analytics

RESULTS

The analysis revealed clear trends in technology usage among developers.

Key findings include:

- The most widely used programming languages.
- The most popular databases among developers.
- Technologies developers plan to learn in the future.

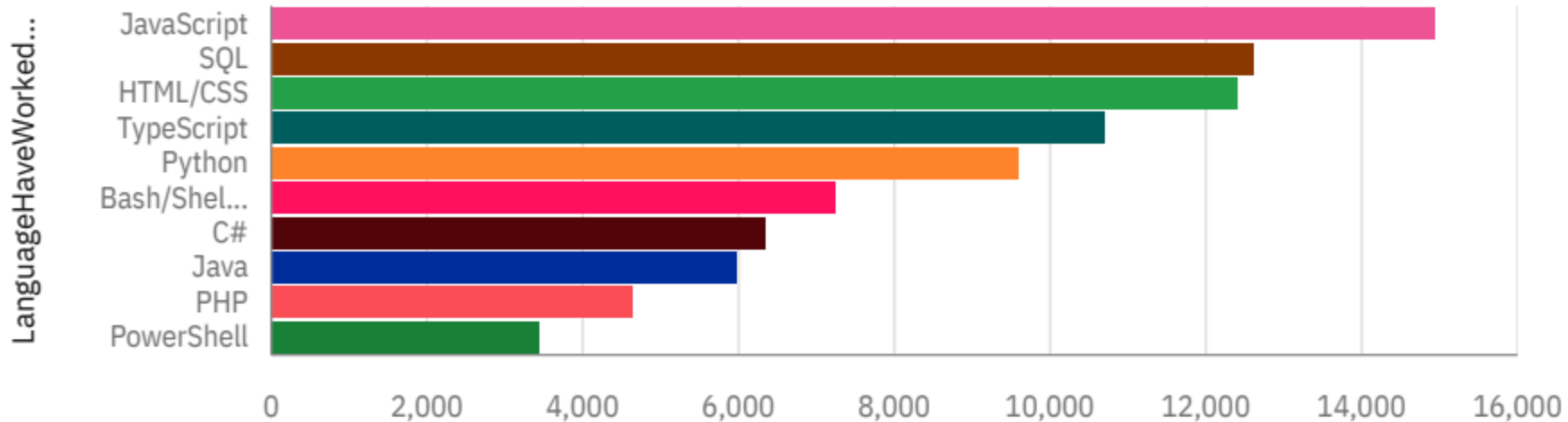
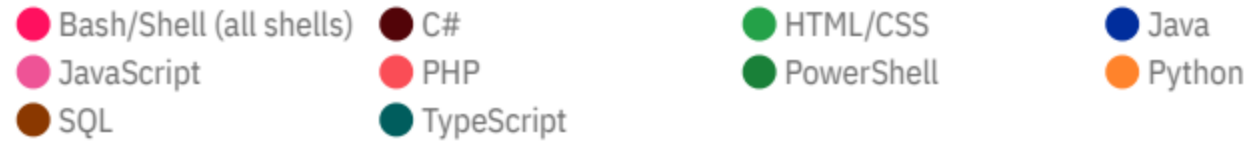
Visualizations were created to illustrate these trends.

PROGRAMMING LANGUAGE TRENDS

Current Year

Top 10 LanguageHaveWorkedWith

LanguageHaveWorkedWith

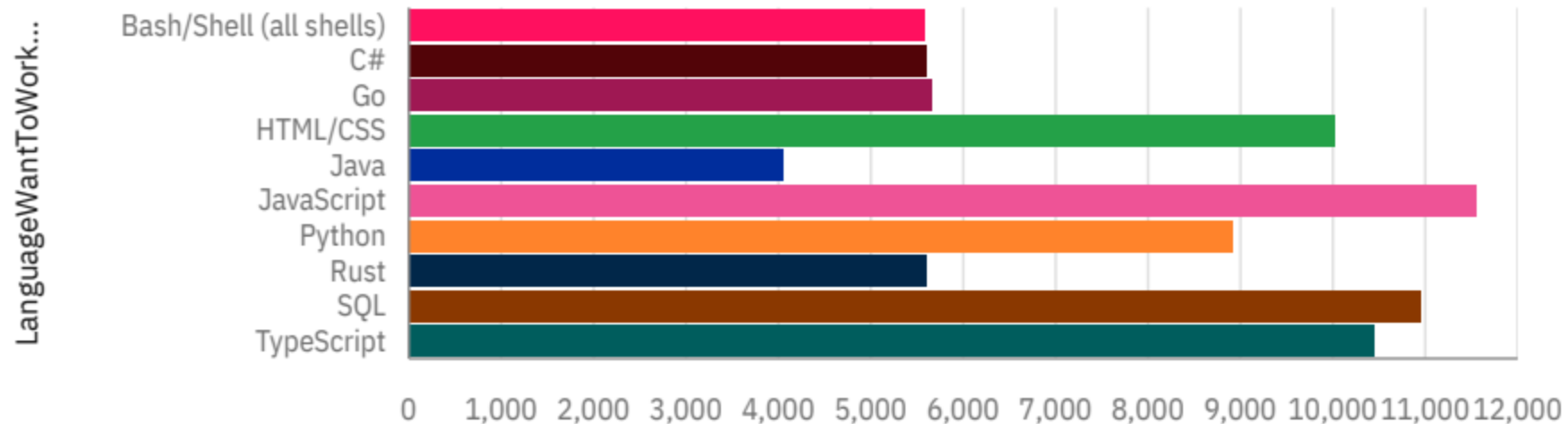


PROGRAMMING LANGUAGE TRENDS

Next Year

Top 10 LanguageWantToWorkWith

LanguageWantToWorkWith



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript and SQL remain dominant programming languages.
- SQL continues to be essential for data-related roles
- Emerging languages such as Rust and Go are gaining popularity.

Implications

- Developers should prioritize learning widely adopted languages
- Companies should focus on hiring skills in high-demand technologies
- Emerging languages may become more important in future development ecosystems



DATABASE TRENDS

Current Year

Top 10 DatabaseHaveWorkedWith

DatabaseHaveWorkedWith

Dynamodb

MongoDB

Redis

Elasticsearch

MySQL

SQLite

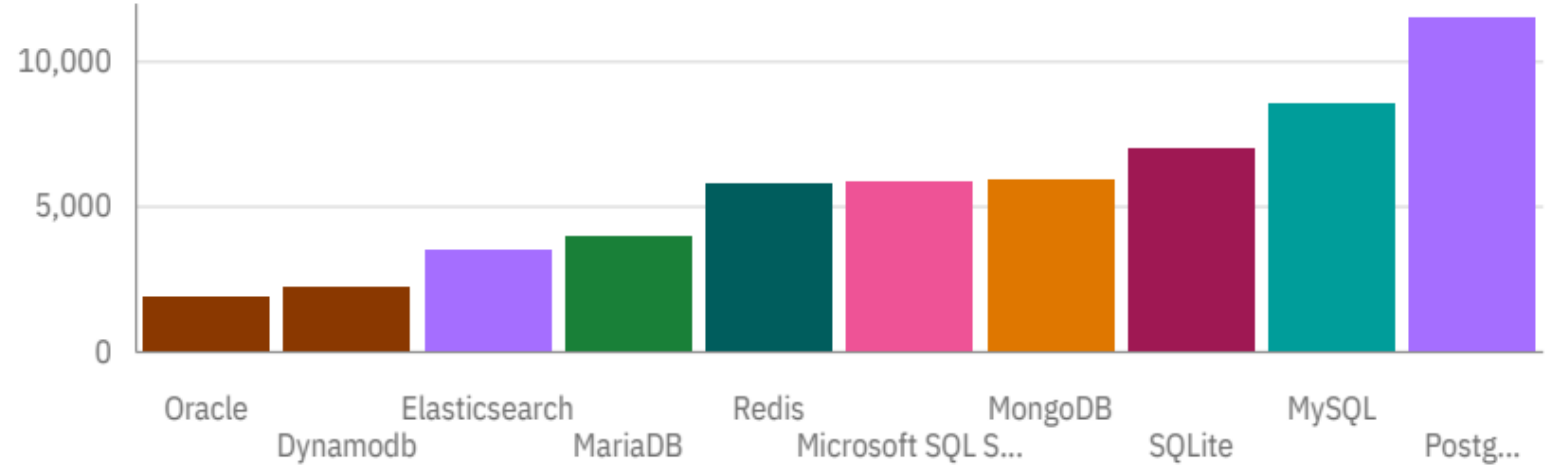
MariaDB

Oracle

Microsoft SQL Server

PostgreSQL

DatabaseHaveWorked...



DatabaseHaveWorkedWith

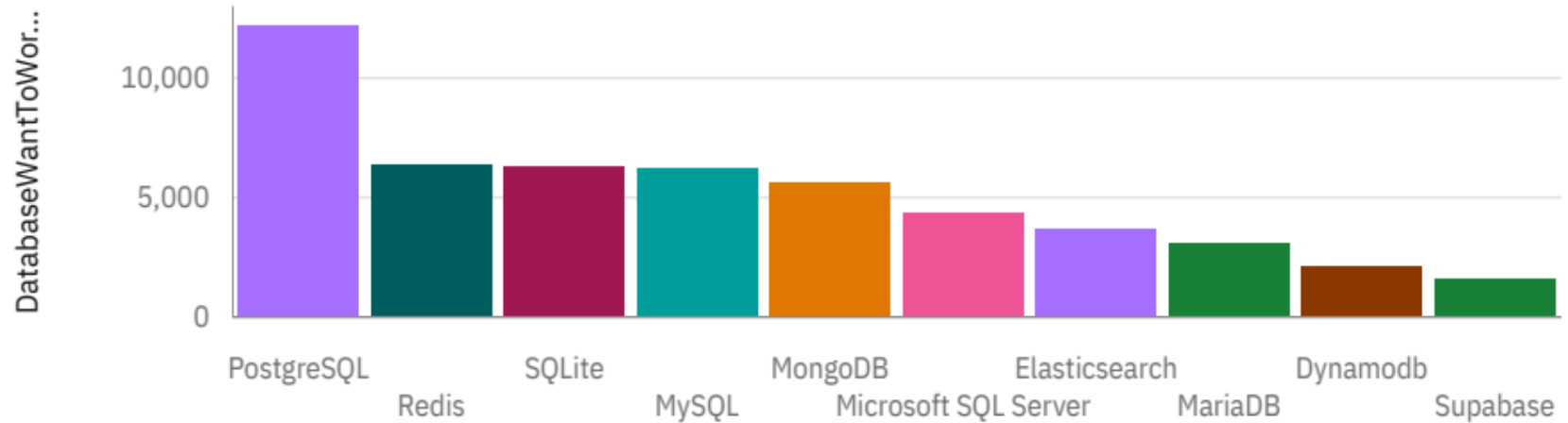


DATABASE TRENDS

Next Year

Top 10 DatabaseWantToWorkWith

DatabaseWantToWorkWith



DatabaseWantToWorkWith



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL and MySQL remain the most widely used databases.
- MongoDB continues to be popular among developers working with NoSQL systems.
- Developers show increasing interest in modern database technologies.

Implications

- Organizations should support both relational and NoSQL databases.
- Database skills remain essential for developers.
- Future technologies may shift toward scalable cloud database systems.



DASHBOARD

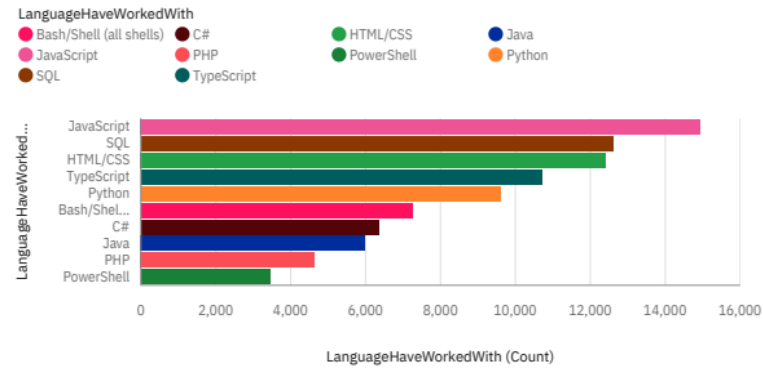


https://github.com/traderedwing/Stack-Overflow-Developer-Survey-Analysis/blob/main/Dashboard_Stack_Overflow.pdf

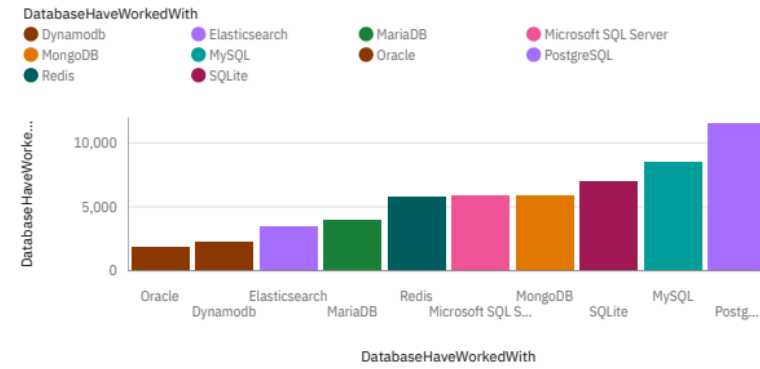
DASHBOARD TAB 1

Current Technology Usage

Top 10 LanguageHaveWorkedWith



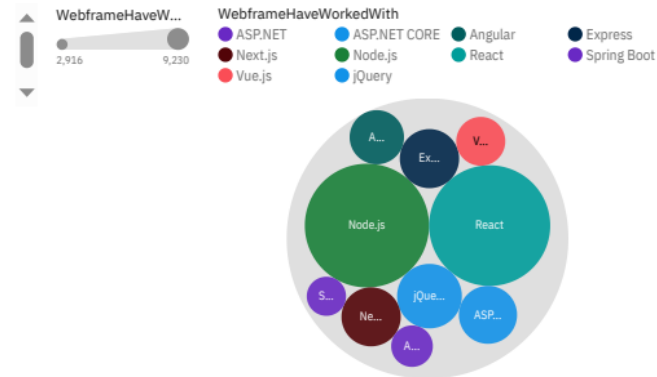
Top 10 DatabaseHaveWorkedWith



Top 10 PlatformHaveWorkedWith



Top 10 WebFrameHaveWorkedWith



Amazon Web Services (AWS)

Google Cloud

Digital Ocean

Cloudflare

Firebase

Heroku

Microsoft Azure

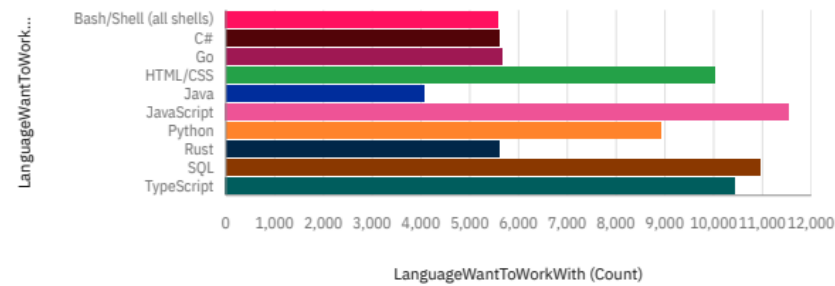


DASHBOARD TAB 2

Future Technology Trend

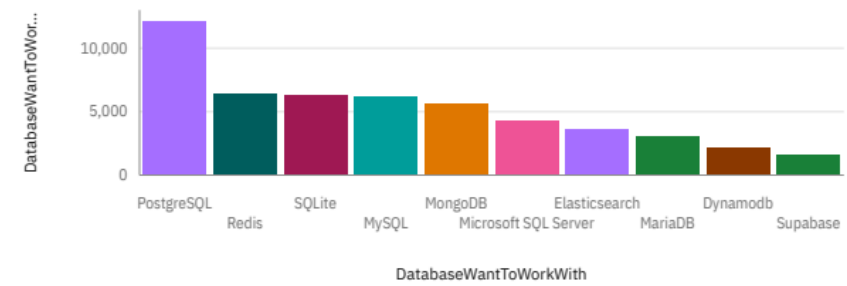
Top 10 LanguageWantToWorkWith

LanguageWantToWorkWith

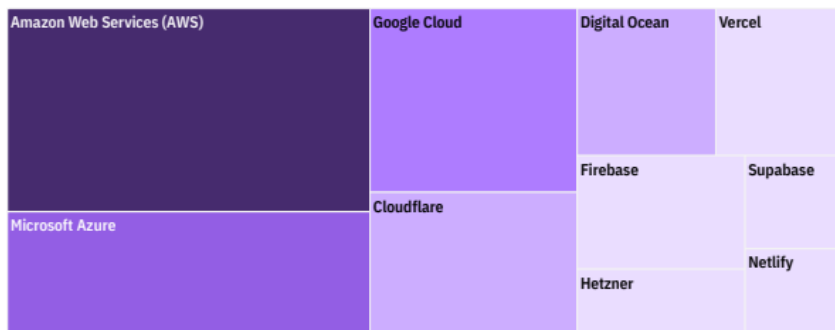
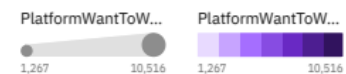


Top 10 DatabaseWantToWorkWith

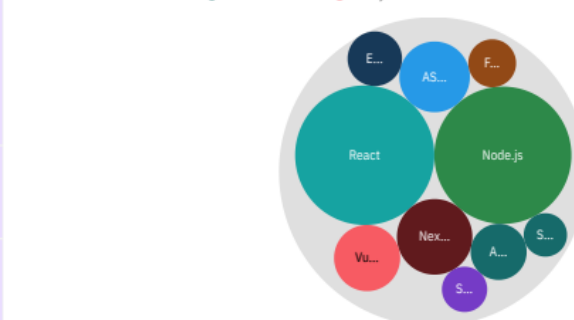
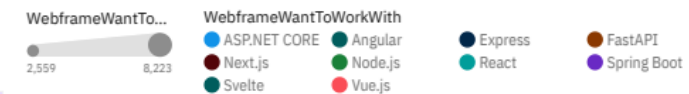
DatabaseWantToWorkWith



Top 10 PlatformWantToWorkWith



Top 10 WebframeWantToWorkWith



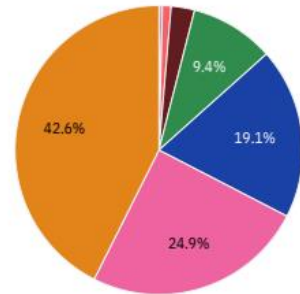
DASHBOARD TAB 3

Demographics

Respondent distribution by Age

Age

- Prefer not to say
- 18-24 years old
- 65 years or older
- 35-44 years old
- Under 18 years old
- 25-34 years old
- 55-64 years old
- 45-54 years old



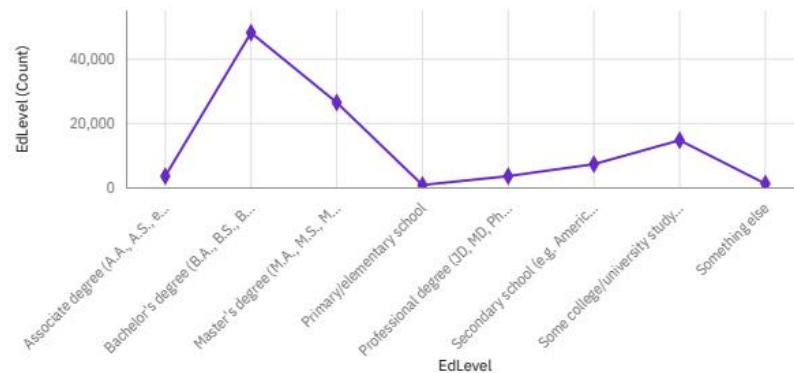
Respondent Count by Country

Country (Count)

1 11,809



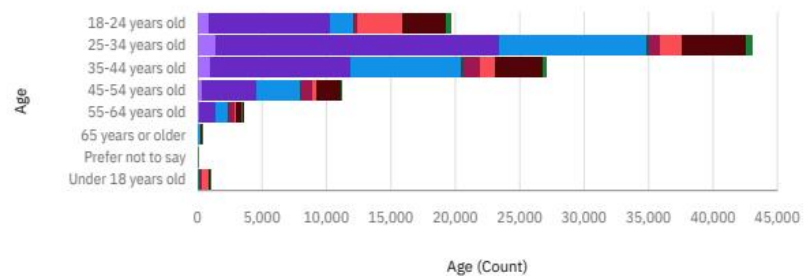
Respondent distribution by Formal Education Level



Age by Age colored by EdLevel

EdLevel

- Associate degree (A.A., A.S., etc.)
- Bachelor's degree (B.A., B.S., B.E...)
- Master's degree (M.A., M.S., M.En...)
- Primary/elementary school
- Professional degree (JD, MD, Ph.D...)
- Secondary school (e.g. American ...)
- Some college/university study wit...
- Something else



DISCUSSION



- The analysis of the Stack Overflow Developer Survey reveals several important technology trends in the software development industry.
- Programming languages such as **JavaScript and SQL** remain dominant due to their versatility and large ecosystems.
- Database technologies such as **PostgreSQL and MySQL** continue to be widely adopted.
- The results also highlight growing interest in **modern and emerging technologies**, which may shape future development trends.
- Data visualization and interactive dashboards help transform raw data into meaningful insights



OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL and HTML/CSS remain the most widely used programming languages among developers.
- Relational databases such as PostgreSQL and MySQL dominate the database landscape.
- Developers are increasingly interested in learning new technologies such as Rust, Go, and modern web frameworks.

Implications

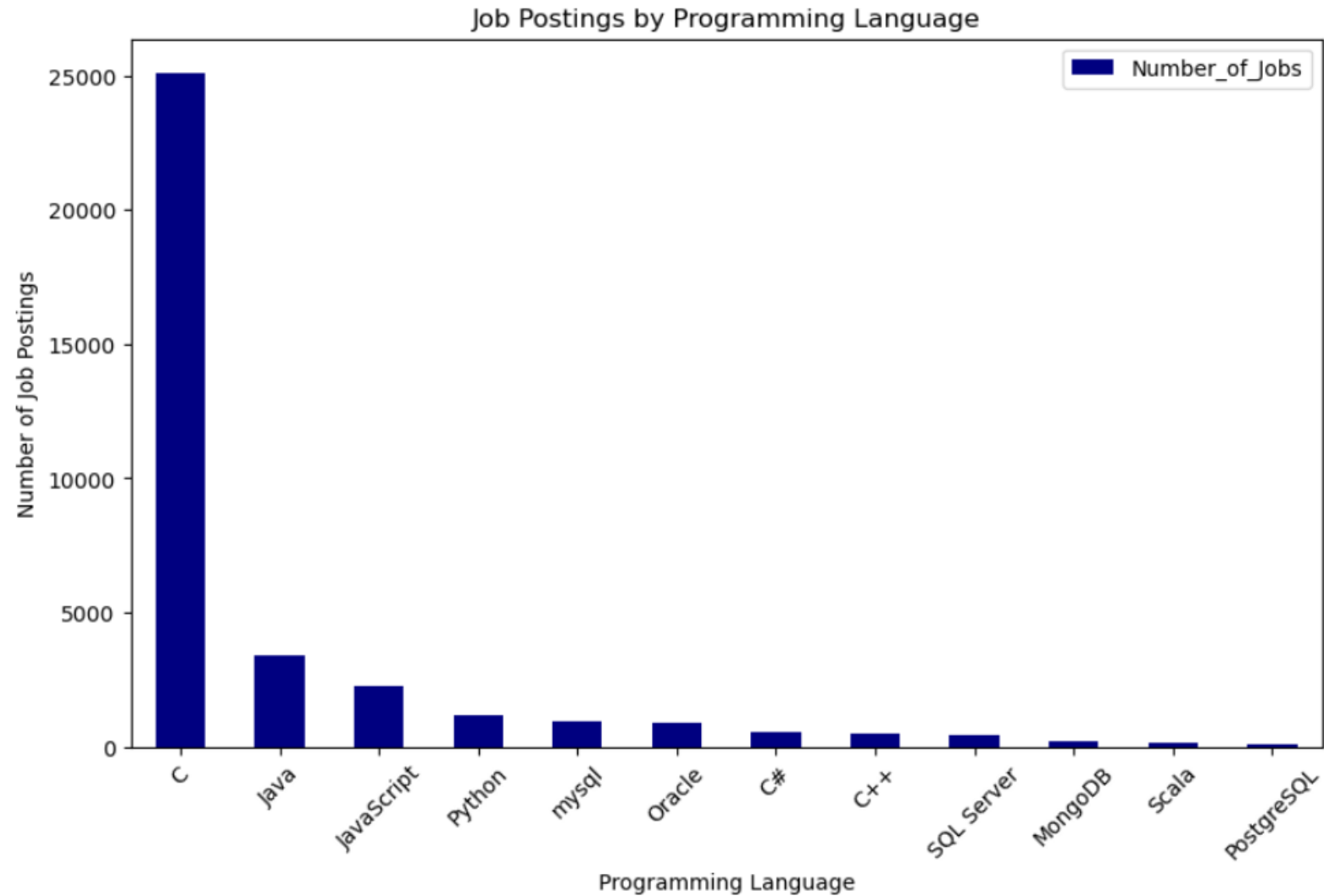
- Developers are increasingly interested in learning new technologies such as Rust, Go, and modern web frameworks.
- Companies should align their hiring strategies with current technology trends.
- Organizations should invest in training programs to support adoption of modern technologies.

CONCLUSION



- The Stack Overflow Developer Survey provides valuable insights into developer technology trends.
- Programming languages such as JavaScript and SQL remain dominant across the industry.
- Database technologies continue to evolve with strong demand for both relational and NoSQL systems.
- Data visualization and dashboards enable effective analysis and communication of technology trends.

JOB POSTINGS



POPULAR LANGUAGES

